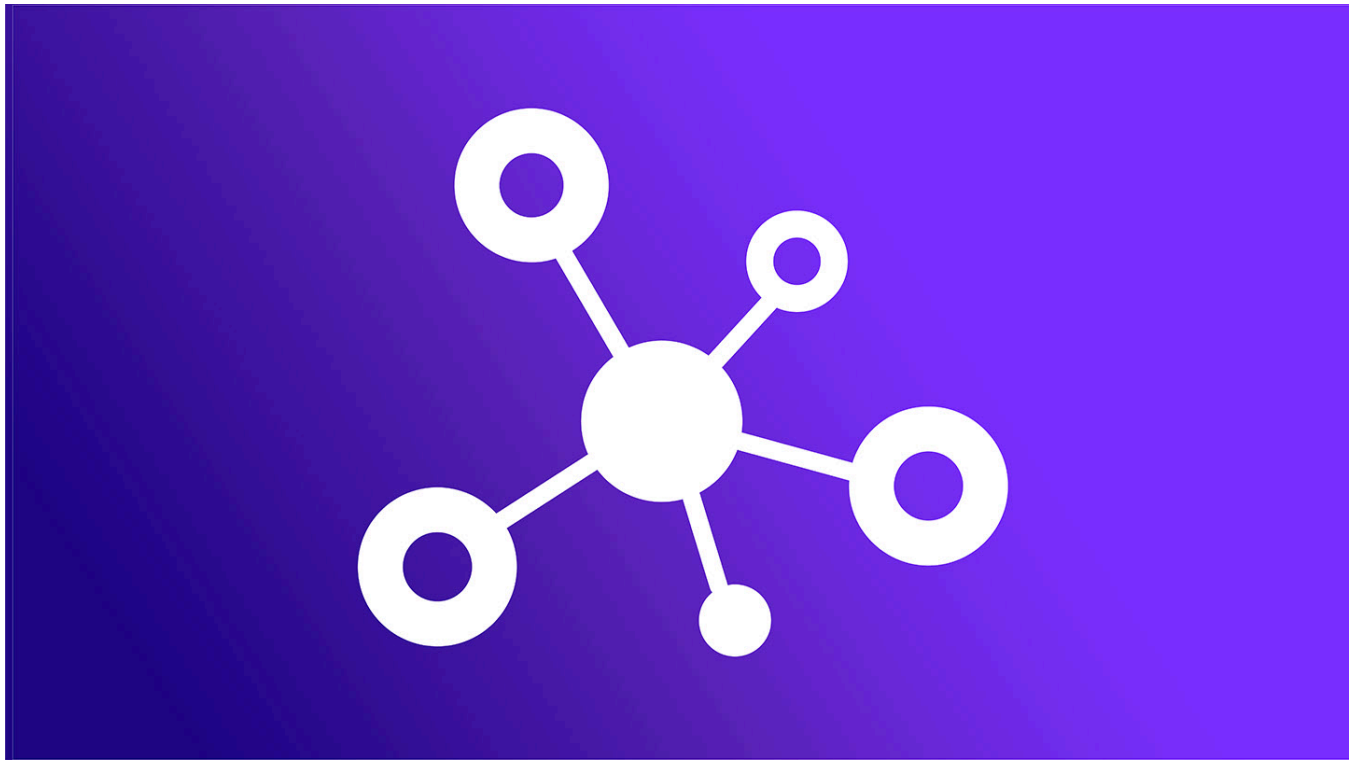
National

Operational Excellence Standard (OES) — Approved cables, connectors, and hardware specifications

HOW22896 | MODIFIED 11/28/2025



Summary

This standard will review approved cables, connectors, and hardware used for

installation of services at the customer's premises.

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What's changed in this article

Date	Summary of change
September 8, 2025	Added to Celestial.

Date	Summary of change
May 1, 2024	Conducted annual review and updated. • Removed high split and added FDX. • Updated terminology to Network Cable. • Updated Cat 5e requirement to Cat 6. • Updated approved fiber connectors. • Added Network feed through connectors. • Added exception for MoCA bond block regarding FDX customers.
March 8, 2022	Conducted scheduled review and updated.
September 30, 2020	Restructured standard to include fiber and twisted pair as well as commercial services standards.
July 30, 2019	Released documents.

At a glance

This article covers the approved cables connectors, and hardware specifications OES.

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General products and supplies

- Only products approved by Comcast Corporate Engineering shall be used in the RF, Category/twisted pair, and/or optical wiring system.
- All optical passive and active devices must meet Comcast NGAN Corporate Engineering guidelines.

Bandwidth frequency ranges for radio frequency (RF) architectures

RF bandwidth

- **54 MHz to 1.2 GHz systems (sub-split):**

    - All passive and active equipment shall support a minimum return RF bandwidth of 5 MHz to 42 MHz.
    - All passive and active equipment must be designed to operate in the 5 MHz to 1.2 GHz frequency range to accommodate MoCA.
- **85 MHz to 1.2 GHz systems (mid-split):**
    - All passive and active equipment shall support a minimum forward RF bandwidth of 85 MHz to 1.2 GHz.
    - All passive and active equipment shall support a minimum return RF bandwidth of 5MHz to 85MHz.
    - All passive and active equipment must be designed to operate in the 5 MHz to 1.2 GHz frequency range to accommodate MoCA.

- **108 MHz to 684 MHz (FDX):**
    - All passive and active equipment shall support a minimum forward RF bandwidth of 108 MHz to 1.2 GHz.
    - All passive and active equipment shall support full duplex bandwidth of 108 to 684 MHz and return path of 5 to 85 MHz.
    - All passive and active equipment must be designed to operate in the 5 MHz to 1.2 GHz frequency range to accommodate full duplex.

Ethernet bandwidth (category and twisted pair)

- Cat5e minimum 1 MHz to 100 MHz.
- Cat6 1 MHz to 550 MHz.

Optical bandwidth

- Drop fiber must pass all required optical wavelengths, CWDM and DWDM.
- Single mode fiber is used for the drop. Wavelengths passed, 1260 nm to 1625 nm.
    - Optical drop passives must pass designated wavelengths in support of the technology used.
    - Optical wavelength filtering will be used in RFoG and EPON networks.

Cable type and specifications

RF cables

- All cables used for RF signal distribution shall meet the minimum physical and electrical requirements of ANSI/SCTE 74 2011, Specification for Braided 75 Ω Flexible RF Coaxial Drop Cable and must be approved by Comcast Corporate Engineering.
- Cables used for RF signal distribution in all new installations must be a minimum size of Series 6 with a minimum foil-braid-foil, tri-shield configuration that include a bonded inner foil tape, outer foil tape bonded to the jacket, and at least 77% braid coverage.
- Cables used for RF distribution shall meet requirements as defined by Comcast Corporate Engineering in relation to LTE spectrum.

Unshielded network pair cable

- All UTP cables used for data distribution shall meet the minimum physical and electrical requirements of ANSI/TIA/EIA-568-B.
- Cables used for data distribution in all new installations must be a minimum of Category 6.

Shielded network pair cable

- STP cables will be found in high ingress potential environments and OSP applications.
- All STP cables used for data distribution shall meet the minimum physical and electrical requirements of ANSI/TIA/EIA-568-B, B2 and B3.
- Cables used for data distribution in all new installations must be a minimum of Category 6.

Fiber optic cable

- All fiber optic cables used for an optical drop shall meet the minimum physical, optical, and electrical requirements of ANSI/ICEA S-110-717-2013 Standard for Optical Fiber

Drop Cable.
- Cables used for an optical drop in all new installations must be pushable, flat, or round depending on drop configuration.
- Optical drop cables must be Comcast XNET Corporate Engineering approved and have a CIFA number.

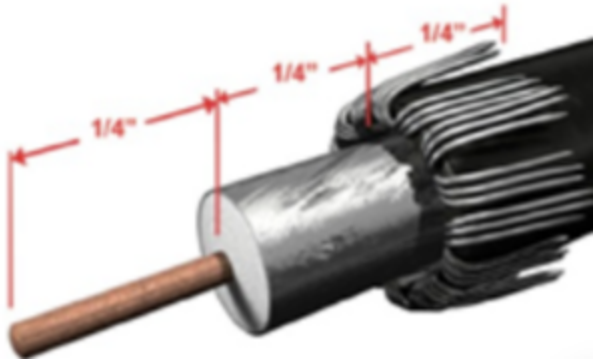
Plenum and riser cables

- Must meet fire safety test standards in accordance with NFPA 262 and outlined in NFPA 90A.

Everything connectors and specifications

F connectors

- All connector installations on existing or new drop cable must be performed only after ensuring the drop is disconnected from live plant to minimize inadvertent injection of noise or other interference into a live network.
- A fixed blade prep tool with a stop must be used. to prep the cable to ¼" x ¼" x ¼" dimensions for Series 6 and ¼, ¼, 5/16 CC for series 11 cables.



- All F connectors must meet the minimum mechanical and electrical specifications of SCTE 123 2011 Specification for "F" Connector, Male, Feed-Through or SCTE 124 2021 for "F" Connector Pin Type.

- All F connectors must meet any additional performance requirements as defined by Comcast Corporate Engineering.
- All F connectors must be compression-type with 360-degree retention of the cable.
- All new outdoor F connectors must have a thread protector attached or pre-installed to the connector.

Tightening of F connectors

- All installed F connectors in the customer's drop system must be tightened with a torque-limiting wrench designed to tighten to 30 inch-pounds. This includes, but not limited to, splitters, traps, locking terminators, and the internal wall side of the wall plates.
- F connectors on all CPE must be securely tightened and not to exceed 10 inch-pounds **except** when a certified prepackaged jumper with a "quick tight" style connector is used.

Weatherproofing

- F connectors with a pre-installed or attached thread protector should be used when possible, instead of a separate thread protector and fitting.
- All outdoor F connector interfaces, including those inside pedestals, equipment enclosures, and any non-climate-controlled environment must be protected to prevent moisture contamination of the threads.
- Silicone compounds must **not** be used to seal or weatherproof outdoor F connector threads.
- Thread protectors must be able to protect port lengths according to SCTE Standard, 0.500".
- Thread protectors that are not permanently installed on to the connector must have a uniform internal inside diameter through the full length of the thread protector. This allows the thread protector to be installed from either direction and ensure a secure compression of the sleeve over a variety of female F-port lengths.

- The F connector must be threaded onto the outdoor F-port and the use of a torque-limiting wrench tightened to 30 inch-pounds to ensure the F-connector pushes against the end of the thread protector for proper sealing.
- Heat-shrinkable tubing must **never** be used on drop cable or F connectors.
- Thread protectors or other sealing devices must **not** be used on indoor connectors or on CPE.

Network wiring connectors

- All network wiring connectors must meet ANSI/TIA-568 specifications.
- Exterior rated connectors or splices must be used in any installation that is not environmentally controlled.



Fiber optic connectors

- All optical connectors must meet minimum mechanical and optical specifications as defined by Comcast Corporate Engineering.
- Optical connector must support an Angel Physical contact (APC) angled end surface or Ultra Physical Contact (UPC) flat end surface.
  - APC is an angled end face used in applications where a high optical return loss (ORL). These are analog optical solutions such as RFoG. To achieve higher ORL an

8-degree angle is polished on the fiber end face.
- APC uses a green connector body.



- UPC uses a polished flat end face across the core. Digital applications work well using a UPC.
- UPC uses a blue connector body.



- Connector types used for optical drops include:
  - SC/APC — RFoG Installations and MetroE
  - SC/UPC — EPON Installations
  - LC/APC — MetroE Installations — utilized on network fiber
- All connector installations on existing or new fiber drop cable must be performed by qualified personnel.
- All fiber handling and fiber safety processes must be observed.

- Drop fiber connectors approved are: pre-connectorized, mechanical connections and fusion.
- All fiber connectors must be secured to the optical tap with a weatherproof environmental protection connector, HFOC.
- All fiber connectors at the optical tap must support APC hardened optical connections.
- All fiber connectors at the NID/NIU should support either SC connectors with APC (RFoG) or UPC (EPON) end faces

Connector preparation tools

RF

- Drop cable preparation tools shall not be capable of field adjustments of the cutting blades.
- Cutting blades in all drop cable preparation tools shall be replaced monthly at a minimum.
- Drop cable preparation tools must have a notch in the dielectric cutting blade to prevent nicking or scoring of the center conductor.
- Cable stops that are a component of any drop cable preparation tools **must not** be removed.
- Compression tools used must match the designed connector style being used.
- Compression tools used to attach fittings to the drop cable shall be inspected monthly and replaced as required.

Network

- Must meet ANSI/TIA-568 specifications for appropriate cable and connector type.
- Feed through connectors are permitted.

Optical fiber

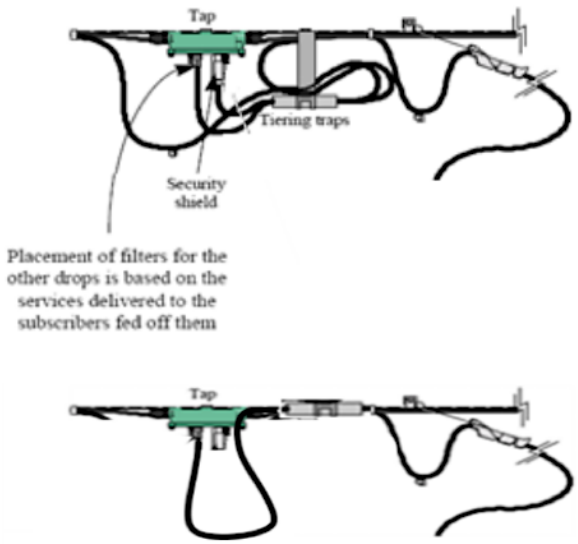
- Qualified personnel must perform fiber splicing or connectorization.

- All splices and connectors must be fusion spliced, approved mechanical connectors or an approved field installable connector.
  - RFOG and Metro E connections must be fusion spliced.
  - EPON connectors can be fusion, mechanical, or field installable.

Hardware for making connections

Filters

- Filters must be weatherproofed with the correct size thread protection on both ends.
- Filters with a pre-installed or attached thread protector should be used instead of a separate thread protector and filter.
- Filters not having a pre-installed or attached thread protector will require a separate thread protector to be installed for use on the SCTE Standard port length of 0.500".
- In an aerial and/or underground installation where available tap port workspace and cable routing does not maintain the minimum cable specifications a jumper between the tap port and trap shall be installed
  - When the trap is attached to a jumper, thread protectors must be used on both ends of the F-81 exterior barrel connector to seal the filter and the F connector.
  - In an aerial installation when a jumper is used, a filter that is installed on a drop must be attached to the strand or to a filter hanger bracket.
  - Filter must be positioned and attached so that there is no strain on the drop and to ensure that the minimum bend radius of the drop cable is always maintained.



RF port termination

- All unused ports must be terminated using terminators with internal 75-ohm resistors.
- Ports that are outside, including those inside pedestals, equipment enclosures, and any non-climate-controlled environment must be protected to prevent moisture contamination of the threads.
- Terminators on tap ports shall have a nonconductive port saver along with thread protection.
- Terminators with a pre-installed or attached thread protector should be used instead of a separate thread protector and terminator.

Barrel connectors

- F-81 drop splice barrel connectors shall meet the minimum requirements of ANSI/SCTE 01 2015, Specification for “F” Port, Female, Outdoor.
- F-81 drop splice barrel connectors must have a non-threaded area at the base of the ports to accommodate weatherproofing devices.
- F-81 indoor drop splice connectors shall meet the minimum requirements of ANSI/SCTE 02 2015, Specification for “F” Port, Female, Indoor.

- F-81 indoor drop splice connectors must have at least one end with threads all the way to the base of the port to accommodate panel nuts and mounting to wall plates.
- F-81 indoor drop splice connectors must never be used in an outdoor application.

Bond blocks

- All bond blocks shall meet the minimum requirements of ANSI/SCTE 129 2017, Drop Passives, Bonding Blocks (Without Surge Protection).
- F-ports must have a non-threaded area at the base of the port to accommodate weatherproofing devices.
- The installation of the MoCA ground block or PoE filter is mandatory on all installations except for FDX customers. See OES MoCA Specification for specific guidance.
- Bond blocks must accommodate wire sizes from # 10 AWG to #12 AWG.

Drop clamps

- Drop clamps must use only the cable's messenger wire to support the drop and must not crush or deform the coaxial cable in any way.
- Drop clamps must be installed to manufacturer specifications
- Drop clamps are to be installed at all locations where tension is introduced onto the cable. (Tap, mid-span, and premise)
- Use of drop clamps at the mid span is optional
- Fiber drop clamps must secure the optical drop per manufacturer and Comcast Engineering specifications.

Fasteners

- Fasteners must not crush, deform, or damage the cable.
- Where applicable, clip type fasteners are preferred over fasteners held in place with screws, nails, or staples.
- Metal staples must not be used on coaxial cables.
- Clip guns (similar to a staple gun) must use a half-round plastic insert to prevent crushing of the RF cable

- Plastic fasteners must be UV and ozone stable.
- Metallic fasteners and metallic components must not rust or otherwise discolor the wood/siding when exposed to the environment.

Splitters and couplers

- All passive devices (splitters, couplers, power inserters) must meet the minimum mechanical and electrical specifications of ANSI/SCTE 153 2016.
- All RF splitters must be MoCA compliant.
- All F-ports must have a non-threaded area at the base of the port to accommodate weatherproofing devices.
- All splitters must meet any additional performance and functional requirements as defined by Comcast Corporate Engineering.
- Optical couplers or splitters are not to be used after the demarcation enclosure.

Drop amplifiers

- Drop amplifiers shall meet the minimum requirements of ANSI/SCTE 161 2016.
- All amplifiers must be single-housing style with integrated RF splitting and RF gain stages within the same housing.
- Amplification on the forward path must provide unity gain.
- All Amplifiers must be MoCA compliant.
- Drop Amplifiers must meet any additional performance and functional requirements as defined by Comcast Corporate Engineering.

ONUs

- All ONUs must meet requirements as defined by Comcast Corporate Engineering.

Wall plates and surface jacks

- Coaxial wall plates must have a 7/16-inch hex recess molded into the wall plate to hold the F-81 and prevent it from turning.
- All Network cable wall plates must meet ANSI/TIA-568 specifications.

- Optical Wall Plates must use a fiber connector bulkhead adapter that supports an SC type connector.
  - Bulkhead adapters will be a snap in style with a shuttered or angled port to restrict exposure to laser light.
  - They must fit NEMA electrical boxes.
- Keystone wall plates must meet the minimum requirements as defined by Comcast Engineering Specifications.

References

- ANSI/SCTE 01 2015, Specification for “F” Port, Female, Outdoor ANSI/SCTE 02 2015, Specification for “F” Port, Female, Indoor
- ANSI/SCTE 74 2011, Specification for Braided 75 Ω Flexible RF Coaxial Drop Cable ANSI/SCTE 123 2011, Specification for “F” Connector, Male Feed-through
- ANSI/ SCTE 124 2021, Specification for “f” Connector, Male Pin Type
- ANSI/SCTE 129 2017, Drop Passives: Bonding Blocks (Without Surge Protection) ANSI/SCTE 153 2016, Drop Passives: Splitters, Couplers and Power Inserters ANSI/SCTE 161 2016, Drop Amplifiers
- ANSI/TIA/EIA-568
- NFPA 70 National Electrical Code NFPA 90A National Electrical Code
- ANSI/SCTE 174 Radio Frequency over Glass Fiber-to-the-Home Specification ITU-T G.652 Single-mode Optical Fiber
- ITU-T G.657 Single-mode Optical Fiber Bend-insensitive Fibers
- ITU-T G.657.B3 Single-mode Optical Fiber Bend-insensitive Fibers (smallest bend radius) ITU-T G987 10G-PON and ITU-T G.988 XG-PON Wavelength Allocation
- IEEE 802.3av 10GEPON PHY parameters

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